Topics 7-8: mechanisms checkpoint

Teacher guide and worked answers

Apply selected ideas about hormonal control, circulation and exchange, then use evidence to choose a precise next step.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C35

Paper 2 topic assessment.

Summer C35 Paper 2 topic-assessment row. This is a teacher-authored 20-mark checkpoint on selected Topics 7-8, not an official Pearson paper, complete Paper 2 coverage or a source of GCSE grade predictions. Core insulin is included; Higher-only glucagon is not required.

Prior learning

Hormones travel in blood to responsive target tissues.

The heart forms two linked circulatory circuits.

Ventilation and diffusion have different mechanisms.

Success evidence

Explain an insulin response without confusing hormone and stored material.

Use vessel direction, pump calculations and exchange mechanisms accurately.

Review item evidence and identify one specific improvement.

Equipment

Shared source page, one selected checkpoint route, feedback page, pen and calculator.

Timer and teacher mark guidance; answers remain with the adult until independent work ends.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Use the 18-minute independent window for Q1-Q5. All three routes carry the same 20 core marks. Record route and any reading, prompting or calculation support.

This is a check after a short recap. Do not present it as an unassisted baseline or use the score to predict a GCSE grade.

Keep teacher answers and live answer-reveal controls out of view during the checkpoint. Reveal feedback only after collecting or pausing responses.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A precise checkpoint, then a useful next step
2	Retrieval	2-4	Retrieve a route and a process
3	I do	4-7	Trace the hormone message
4	I do	7-9	Read direction before oxygen content
5	I do	9-11	Volume and pressure explain ventilation
6	We do	11-13	Use an analogous calculation
7	We do	13-15	Challenge a pair of claims
8	We do	15-16	State what the calculation cannot show
9	Hinge	16-17	Before you begin
10	You do	17-18	Set up your selected checkpoint route
11	You do	18-36	Independent checkpoint: 18 minutes
12	Review	36-38	Feedback starts now
13	Review	38-39	Audience: identify the next move
14	Exit	39-40	Finish with a targeted action
15	Exit	Within stage	Store the evidence, not a label

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A precise checkpoint, then a useful next step

Explain that this is original selected-topic practice after a recap. Record the chosen route and support. No grade prediction or public score ranking.

2 Retrieval Retrieve a route and a process

R1 blood. R2 ventilation moves air through pressure changes; diffusion is net particle movement down a gradient. Accept spoken or pointed responses.

3 I do Trace the hormone message

Think aloud through location → carrier → response. This general recap does not rehearse the checkpoint wording. Recognition explains why every cell does not respond identically.

4 I do Read direction before oxygen content

Trace right ventricle → pulmonary artery → lungs. It carries relatively oxygen-poor blood away from the heart. Let pupils see how the vessel definition differs from its oxygen content.

5 I do Volume and pressure explain ventilation

Trace one mechanical sequence. Distinguish the resulting airflow from diffusion across the alveolar barrier. This is a recap, not a breathing practical.

6 We do Use an analogous calculation

Model M = 3,000 mL/min = 3 L/min. Joint N = 4,000 mL/min = 4 L/min. Cancel beat units, then divide by 1,000. These are warm-up records, not checkpoint A/B.

7 We do Challenge a pair of claims

W2 false: responsiveness/recognition matters. W3 false: it leaves the heart, so it is an artery. Pupils justify with a route rather than merely voting.

8 We do State what the calculation cannot show

No. These are constructed records; health, demand and oxygen content are not measured. Keep the conclusion to the supplied output comparison.

9 Hinge Before you begin

Correct: answer independently with agreed support recorded; review afterwards. Read the support record aloud if needed.

A. Record the route and support; answer first; use feedback afterwards. - That makes the evidence interpretable and keeps feedback separate from the attempt.

B. Convert the total directly into a predicted GCSE grade. - Twenty marks of selected teacher-authored content cannot establish an official grade.

C. Copy the worked answers while completing the checkpoint. - That would hide which ideas the pupil can currently apply.

10 You do Set up your selected checkpoint route

Explain calculator and read-aloud access. Write the route/support on the feedback page before timing begins. Do not impose a penalty for the route choice.

11 You do Independent checkpoint: 18 minutes

Keep answer reveals closed. Offer agreed access support without supplying scientific content; record any additional content prompts. Optional challenge audit is unscored and only after the core.

12 Review Feedback starts now

Use the five four-mark item guides. A pupil can mark one selected item while teacher completes the rest after class; do not try to read every model answer aloud in two minutes. Preserve the original response before correction.

13 Review Audience: identify the next move

Accept a quiet conference. Convert “revise everything” into a specific action, such as pressure direction, mL-to-L conversion or hormone versus storage.

14 Exit Finish with a targeted action

No new scored science items. Use Q1-Q5 evidence to justify the chosen practice; avoid labelling a pupil by the score.

15 Exit Store the evidence, not a label

Shared exit minute. Retest with a fresh equivalent item later if useful; do not reuse the same answer as independent evidence.

Answers and assessment

Retrieval, We do and hinge

R1 blood. R2 ventilation moves air; diffusion is net particle movement down a concentration gradient. W1 M = 3,000 mL/min = 3 L/min; N = 4,000 mL/min = 4 L/min. W2 target-cell recognition/responsiveness matters. W3 pulmonary artery is an artery because it leaves the heart. Output alone does not establish health. Correct hinge: record support, attempt first and review afterwards.

Q1: 4 marks, same across all routes

Award one mark each: pancreas [1]; insulin [1]; one valid response such as increased glucose uptake in responsive tissues or increased glycogen storage in liver/muscle [1]; blood glucose decreases towards its usual range [1]. Do not require beta-cell terminology. Do not credit insulin becoming glycogen. Some glucose uptake is insulin-independent.

Q2: 4 marks, same across all routes

Lungs → left atrium [1] → left ventricle [1] → aorta [1] → body. Artery carries blood away from the heart [1]. Accept LA/LV if meaning is clear. Oxygen-rich is not a sufficient definition; pulmonary artery is the counterexample.

Q3: 4 marks, same across all routes

A: 75 × 80 [1] = 6,000 mL/min [1]; 6 L/min [1]; output also depends on stroke volume, so heart rate alone is insufficient [1]. Accept consistent error-carried-forward for the conversion method if a previous arithmetic slip occurred. If B is used as evidence: 100 × 50 = 5,000 mL/min = 5 L/min, despite its higher heart rate. A correct unsupported final 6 L/min can receive the result and conversion marks, but not the explicit-working mark.

Q4: 4 marks, same across all routes

Thoracic volume increases [1]; pressure falls below atmospheric/surrounding air pressure [1]; oxygen diffuses from alveolar air into arriving blood down its concentration gradient [1]; a thin barrier shortens diffusion distance [1]. For the third mark require direction and gradient. Cellular respiration and ventilation alone do not name the exchange mechanism.

Q5: 4 marks, same across all routes

Thickness changes, from 1 to 2 [1]. V = 4 × 3 / 2 = 6 [1]. U = 12, so V has half U’s index [1]. One specific limit [1]: relative uncalibrated inputs, omitted proportionality/diffusion coefficient, uniform fixed barrier/concentration difference, no changing blood flow or ventilation, or no tissue-demand information. Do not accept only “it is a model” without explaining the limitation. Allow consistent comparison reasoning after an arithmetic slip; do not award two marks for one unexplained value.

Optional unscored audit

A defensible answer identifies oxygen content, distribution to the relevant tissue, exchange effectiveness and/or tissue oxygen demand as additional evidence. Explain why a volume per minute alone cannot show oxygen adequacy. Do not use this answer to add marks to the 20-mark core.

Marking and feedback

Five items × four marks = 20. Record route and support separately from marks; scores from different support conditions are not directly interchangeable. This small checkpoint has no official grade boundaries. Use item evidence: Q1 signal/response reteach; Q2 trace left-heart route; Q3 units and multiplication; Q4 volume/pressure and gradient; Q5 controlled factors and model scope.

L1 and exit

Accept a specific correction tied to an item and a scientific reason. E1 names a demonstrable idea; E2 identifies a concrete practice, such as a fresh mL-to-L calculation or annotating a pressure sequence. Generic confidence alone is not the intended evidence.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

Assessment sequence: analogous recap and joint examples, then 18 uninterrupted minutes for the selected 20-mark route. Teacher feedback begins only after the first attempt is paused or collected.

The interactive panel uses only warm-up cases M/N and general route principles. It does not reveal checkpoint A/B output, U/V calculations or the Q1-Q5 mark scheme.

Give focused feedback to one chosen item during the three-minute review and complete any remaining marking afterwards. Preserve original responses and corrections as separate evidence.

Same core concepts and marks across routes. The supported word bank and reading support affect interpretation, so record them; do not assign a fixed attainment identity from a route or score.

Access and responsive teaching

Supported

Same five core questions and marks, with chunked wording, a word bank and optional read-aloud. Record these supports alongside the result.

Standard

Same five core questions and marks with independent explanations. Calculator access is allowed and recorded.

Stretch

Same 20-mark core first. An optional unscored model-audit question follows if time permits; it does not change the score.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

The hormone insulin becomes the glycogen store.

Glucose is used to build glycogen; insulin is a signal.

Check: Use Q1 to distinguish signal from response.

An artery is defined by oxygen content.

An artery carries blood away from the heart, including the pulmonary artery.

Check: Use Q2 vessel-direction explanation.

Inhalation begins because lung pressure rises.

Increasing thoracic volume lowers pressure below the surroundings, drawing air in.

Check: Use Q4 pressure direction.

A checkpoint total is a GCSE grade.

This small teacher-authored check samples selected content and records support.

Check: Use item patterns to choose teaching, not a grade conversion.

Word	Meaning
homeostasis	Regulation of internal conditions within limits.
insulin	A pancreatic hormone involved in reducing raised blood glucose.
artery	A vessel that carries blood away from the heart.
cardiac output	Volume pumped by one ventricle per unit time.
ventilation	Movement of air into and out of the lungs.
diffusion	Net movement of particles down a concentration gradient.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

NIDDK: What is diabetes?

Checked core insulin and glucose roles. Checked 9 September 2026.

NHLBI: How blood flows through the heart

Checked chamber/vessel routes and valve direction. Checked 9 September 2026.

NHLBI: How the lungs work

Checked ventilation and exchange context. All checkpoint items and model data are original. Checked 9 September 2026.